1) What is HTML ?

- HTML stands for HyperText Markup Language. It is the standard markup language used to create web pages.
- HTML is a combination of Hypertext and Markup language.
- Hypertext defines the link between web pages. A markup language is used to define the text document within the tag to define the structure of web pages.
- It uses HTML tags and attributes to describe the structure and formatting of a web page
- HTML consists of various elements, that are responsible for telling search engines how to display page content. For example, headings, lists, images, links, and more.

2) Tell me the basic structure of the html page?

- The basic structure of an HTML page contains the essential building-block elements (i.e. doctype declaration, HTML, head, title, and body elements) upon which all web pages are created.

3) who has created HTML ?

- HTML was created by Tim Berners-Lee in 1991.
- The first-ever version of HTML was HTML 1.0, but the first standard version was HTML 2.0, published in 1995.
- Currently, we are using HTML5, which is the latest and most recent version of HTML.

4) What are the advantages of using the html?
Advantages:

- Ease of Use: HTML is relatively easy to learn and understand, making it accessible to beginners.
- Versatility: HTML can be used to create a wide variety of content, from simple web pages to complex multimedia presentations.
- Compatibility: HTML is supported by all major web browsers, ensuring that content displays consistently across different platforms and devices.
- Integration with Other Technologies: HTML seamlessly integrates with other web technologies such as CSS for styling and JavaScript for interactivity, allowing for rich and dynamic web experiences.
- Accessibility: HTML provides features and attributes that support accessibility, making it possible to create web content that is usable by people with disabilities.

5) what are the disadvantages of html?
Disadvantages:

- Limited Styling Options:
  - HTML alone has limited styling capabilities, often requiring additional technologies like CSS to achieve more complex designs.
- Semantic Markup Complexity:
  - Creating semantically meaningful HTML code requires careful consideration and understanding of best practices, which can be challenging for beginners.
- Browser Compatibility Issues:
  - While HTML itself is widely supported, different browsers may interpret HTML code differently, leading to compatibility issues that require extra testing and troubleshooting.
- Security Concerns:

- HTML is vulnerable to various security threats such as cross-site scripting (XSS) and injection attacks, requiring developers to implement proper security measures to protect against these risks.

- Maintenance Overhead:
  - Large and complex HTML documents can be difficult to maintain and update, especially without proper organization and documentation.

6) What are HTML Entities?

- In HTML some characters are reserved like '<', '>', '&', etc. To use these characters in our webpage we need to use the character entities called HTML Entities.
- For eg: < > and &

7) What are the various formatting tags in HTML?
Various formatting tags are strong,em,ins,strike,big,small tags etc;

8) What is the difference between , <b> tags and , <i> tags?

- The effect on a normal webpage of the tags , <b> and , <i> is the same. <b> and <i> tags stands for bold and italic.
- These two tags only apply font styling and bold tag <b>, just adds more ink to the text, these tags don't say anything about the text.
- Whereas, and tags represent that the span of text is of strong importance or more importance and emphatic stress respectively than the rest of the text.
- These tags have semantic meaning.

9) What is the significance of <head> and <body> tag in HTML?

- <head> tag provides the information about the document.

- It should always be enclosed in the <html> tag. This tag contains the metadata about the webpage and the tags which are enclosed by head tag like <link>, <meta>, <style>, <script>, etc. are not displayed on the web page.

- Also, there can be only 1 <head> tag in the entire Html document and will always be before the <body> tag.

- <body> tag defines the body of the HTML document. It should always be enclosed in the <html> tag. All the contents which needs to be displayed on the web page like images, text, audio, video, contents, using elements like <p>, <img>, <audio>, <heading>, <video>, , etc. will always be enclosed by the <body> tag.

- Also, there can be only 1 body element in an HTML document and will always be after the <head> tag.

10) How is Cell Padding different from Cell Spacing?

- Cell Spacing is the space or gap between two consecutive cells.
- Whereas, Cell Padding is the space or gap between the text/ content of the cell and the edge/ border of the cell.

11) What is the difference between "display: none" and "visibility: hidden", when used as attributes to the HTML element?

- When we use the attribute "visibility: hidden" for an HTML element then that element will be hidden from the webpage but still takes up space.

- Whereas, if we use the “display: none” attribute for an HTML element then the element will be hidden, and also it won’t take up any space on the webpage.

12) Difference between link tag <link> and anchor tag <a>?

- The anchor tag <a> is used to create a hyperlink to another webpage or to a certain part of the webpage and these links are clickable, whereas, link tag <link> defines a link between a document and an external resource and these are not clickable.

14) What are Semantic Elements?

- Semantic elements are those which describe the particular meaning to the browser and the developer. Elements like <form>, <table>, <article>, <figure>, etc., are semantic elements.

15) Define Image Map?

- Image Map lets a developer map/link different parts of images with the different web pages.
- It can be achieved by the <map> tag in HTML5, using which we can link images with clickable areas.

```
<img src=”image_url” , usemap=”#workspace” />
<map name=”workspace”>
     <area shape=”rect” coords=”34, 44, 270, 350” , href=”xyz.html” />
     <area shape=”rect” coords=”10, 120, 250, 360” , href=”xyz.html” />
</map>
```

16) What are some of the advantages of HTML5 over its previous versions?

- Some advantages of HTML5 are:-
- It has Multimedia Support.
- It has the capabilities to store offline data using application cache.
- Javascript can be run in the background.
- HTML5 also allows users to draw various shapes like rectangles, circles, triangles, etc via scripting and canvas tag.
- Included new Semantic tags and form control tags.

17) Explain the concept of web storage in HTML5.

This web storage helps in storing some of the static data in the local storage of the browser so that we do not need to fetch it from the server every time we need it.

- There is a size limit based on different browsers.
- This helps in decreasing the load time and a smooth user experience.
- There are two types of web storage that are used to store data locally in HTML5:
    - Local Storage - This helps in storing data that will be retained even though the user reopens the browser. It is stored for each webapp on different browsers.
    - Session Storage - This is used for one session only. After the user closes the browser this gets deleted.

18) What are the correct extensions for saving an HTML file?

- correct extensions for saving an HTML file is .html or .htm

19) What is the difference between an absolute and relative URL?

- An absolute URL includes the full web address, the protocol (such as http or https) and the domain name (such as www.example.com).
- A relative URL, on the other hand, specifies the location of a resource relative to the current web page.

For example, a relative URL might include the file path (such as /images/picture.jpg) or the relative path (such as ../images/picture.jpg).

20) What is the role of the alt attribute in HTML?

- The alt attribute provides alternative text for an image in case the image cannot be displayed.
- This is important for accessibility, as screen readers can read the alt text to describe the image to visually impaired users.

21) What is the role of the title attribute in HTML?

- The title attribute provides additional information about an element, such as a link or an image.
- The title text is displayed as a tooltip when a user hovers over the element.

22) What is a form in HTML?

- A form is a set of input fields and other elements to collect user data.
- Forms can be used for various purposes, such as logging in, submitting feedback, or purchasing.

23) What is the role of the action attribute in HTML forms?

- The action attribute is used to specify the URL of the script or program that will process the data submitted by the form.
- When the user clicks the submit button, the form data is sent to the specified URL for processing.

24) What is the role of the method attribute in HTML forms?

- The method attribute is used to specify the HTTP method that will be used to submit the form data.
- The two most common methods are GET and POST.

- GET is used to retrieve data from the server, while POST is used to send data to the server

25) In how many ways can you display HTML elements?

- HTML elements can be displayed in several ways, including block, inline, inline-block, and none. The display property can specify how an element should be displayed.

26) How to handle events in HTML?

- Events can be handled using JavaScript, which can be included in the HTML document using the <script> tag.
- Event listeners can be added to HTML elements using the addEventListener() method, which allows custom code to be executed in response to user actions, such as clicks or keystrokes.

27) Can we display a web page inside a web page or Is nesting of webpages possible?

- Yes, we can display a web page inside another HTML web page.

- HTML provides a tag <iframe> using which we can achieve this functionality.

<iframe src="url of the web page to embed" />

28) What is the purpose of the "z-index" property in CSS?

- The "z-index" property specifies the stack order of an element. An element with a higher "z-index" value will appear above elements with lower values.

29) Explain the box model in CSS.

- The box model refers to the layout design of elements in CSS, which consists of content, padding, border, and margin.

30) What is the difference between "padding" and "margin" in CSS?

- Padding is the space between the content and the border, while margin is the space outside the border.

31) What is the CSS float property used for?

- The float property is used to align elements horizontally within their container.

32) What is the purpose of the "clearfix" hack in CSS?

- The clearfix hack is used to contain floated elements within a container, preventing it from collapsing.

33) What are media queries in CSS used for?

- Media queries are used to apply different styles based on the characteristics of the device or browser, such as screen width, height, and orientation.

34) What is the "box-sizing" property in CSS used for?

- The "box-sizing" property is used to specify how the total width and height of an element are calculated, including padding and border.

35) What is the difference between "em" and "rem" units in CSS?

- "em" units are relative to the font size of the parent element.
- while "rem" units are relative to the font size of the root element (typically the <html> element).

36) What is the purpose of the "transition" property in CSS?

- The "transition" property is used to create smooth transitions between different property values over a specified duration.

37) What is the CSS "display" property used for?

- The "display" property specifies the rendering behavior of an element, such as block, inline, inline-block, none, etc.

38) Explain the concept of specificity in CSS?

- Specificity determines which CSS rule is applied to an element when multiple rules conflict. It is calculated based on the combination of selectors used to target an element.

39) What is the purpose of the "min-width" and "max-width" properties in CSS?

- "min-width" sets the minimum width for an element, ensuring it never becomes smaller than the specified value.
- "max-width" sets the maximum width for an element, preventing it from becoming larger than the specified value.

40) What is the difference between "px", "em", and "rem" units in CSS?

- "px" units are fixed-size units, "em" units are relative to the font size of the parent element, and "rem" units are relative to the font size of the root element.

41) What is the purpose of the "flex" property in CSS?

- The "flex" property is used in flexbox layouts to specify how a flex item grows or shrinks to fill the available space.

42) How do you create a CSS animation?

- By using the "@keyframes" rule to define the animation's keyframes and then applying the animation to an element using the "animation" property.

43) What is the purpose of the "transform" property in CSS?

- The "transform" property is used to apply transformations such as scaling, rotating, skewing, or translating an element in.

44) What is the difference between CSS grid and CSS flexbox?

- CSS grid is a two-dimensional layout system for arranging elements in rows and columns.
- while CSS flexbox is a one-dimensional layout system for arranging elements in a single row or column.

45) What is the difference between pseudo-classes and pseudo-elements?

- Pseudo-classes select elements based on their state, such as :hover or :active, while pseudo-elements select parts of an element, such as ::before or ::after.

46) How do you override CSS styles?

- Use more specific selectors, use !important, or rearrange the order of stylesheets to override CSS styles

47) What is the difference between overflow: hidden and overflow: auto in CSS?

- overflow: hidden hides any content that overflows the element's box.
- while overflow: auto adds scrollbars to the element if its content overflows.

48) What is the purpose of the opacity property in CSS?

- The opacity property specifies the transparency level of an element, allowing it to appear partially transparent.

49) How do you target an element with multiple classes in CSS?

- .class1.class2 { /* styles */ }.

50) What is the purpose of the word-wrap property in CSS?

- The word-wrap property in CSS is used to control how long words are handled when they exceed the width of an element, allowing you to wrap long words onto the next line or break them at specific characters.
- For e.g: word-wrap: break-word;

51) What is responsive web design?

- Responsive design is an approach to web page creation that makes use of flexible layouts, flexible images and cascading style sheet media queries.
- The goal of responsive design is to build web pages that detect the visitor’s screen size and orientation and change the layout accordingly.